

Secure Browser-Native Remote Assistance & IT Diagnostics

White Paper

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ShivTrix Admin v4.2 PRO - remote control, hardware telemetry and IT diagnostics

1. Executive summary

ShivTrix Admin is a browser-native remote-assistance and IT-diagnostics suite. A support technician can see a computer's screen, operate its mouse and keyboard with the same ease as a desktop remote-access tool - from a laptop *or a touch-screen phone* - and run network diagnostics (ping, traceroute, DNS lookup, port check, adapter inventory) **on the remote computer itself**, all without the technician installing anything. The host runs one small local companion program; the technician needs only a browser, the installable PWA, or the Android app.

This paper explains why the product is built the way it is, how a session works end to end, which security controls protect the person whose computer is being operated, how the system was verified, and - just as important - what it does not do yet.

Capability	How it is delivered
Screen sharing	WebRTC, peer to peer, encrypted by the browser (DTLS-SRTP)
Mouse, touch and keyboard control	Normalised input events over a WebRTC data channel, injected by the companion with the Windows SendInput API
Diagnostics on the host	Allow-listed network tools executed by the host's companion, streamed back over the data channel
Group sessions	Up to 10 technicians per room; one active controller at a time
Platforms	Windows host (EXE or Python); technician on any modern browser, installable PWA, Android app
Look and feel	OLED-black "Pro" interface with the ShivTrix blue-and-gold identity

> **Design principle**
> The person sitting at the computer is always in control of the session. Control is never granted by a web page: it requires a native Windows confirmation dialog that only a person at that PC can answer, and it can be withdrawn at any moment.

2. Problem and goals

Support teams, IT administrators and family "tech helpers" repeatedly face the same trade-offs with conventional remote-desktop software:

- **Install friction.** The person who needs help must download, install and sometimes register an account before any help can begin.
- **Touch is an afterthought.** Operating a desktop from a phone is clumsy unless the product models touch gestures properly.
- **Diagnostics run on the wrong machine.** A technician's own ping tells them about *their* network, not the customer's.
- **Opaque trust.** It is often unclear who may control the machine, for how long, and what was done.

ShivTrix Admin was designed against five goals: (1) zero-install for the technician, (2) a single lightweight component on the host, (3) first-class touch and keyboard control, (4) host-side diagnostics, and (5) explicit, revocable, auditable consent.

3. Solution overview

The suite has three layers. Each can be used on its own.

Layer	Role	Notes
ShivTrix Admin web app / PWA	User interface for both host and technician: dashboards, Remote Resolve studio, Power Toys, IT toolkit	Static files; works from GitHub Pages or the companion; installable and offline-capable
Companion	Serves the app, exposes telemetry and diagnostics, runs the remote-control engine	Python program or packaged EXE; listens on 127.0.0.1 only by default
Platform wrappers	Windows EXE/installer, Android APK, installable PWA	Built from the same web files; no separate code base

3.1 Remote Resolve studio

The host shares the **entire screen** and chooses an access level. The technician joins with a link, QR code or room ID. Three levels exist, and only the host can change them:

Level	Technician can
View Only	Watch the shared screen
Laser Pointer	Point at items on the host screen
Assisted Control	Use mouse, touch and keyboard on the host and run Power Toys on the host system

3.2 Power Toys on the host system

In Assisted Control a technician can switch the diagnostics panel from **This device** to **Host system**. Ping, continuous ping, traceroute, NSLookup, port check and the network-adapter inventory then execute on the host and stream results back live, so the technician sees the customer's real network conditions.

4. Architecture

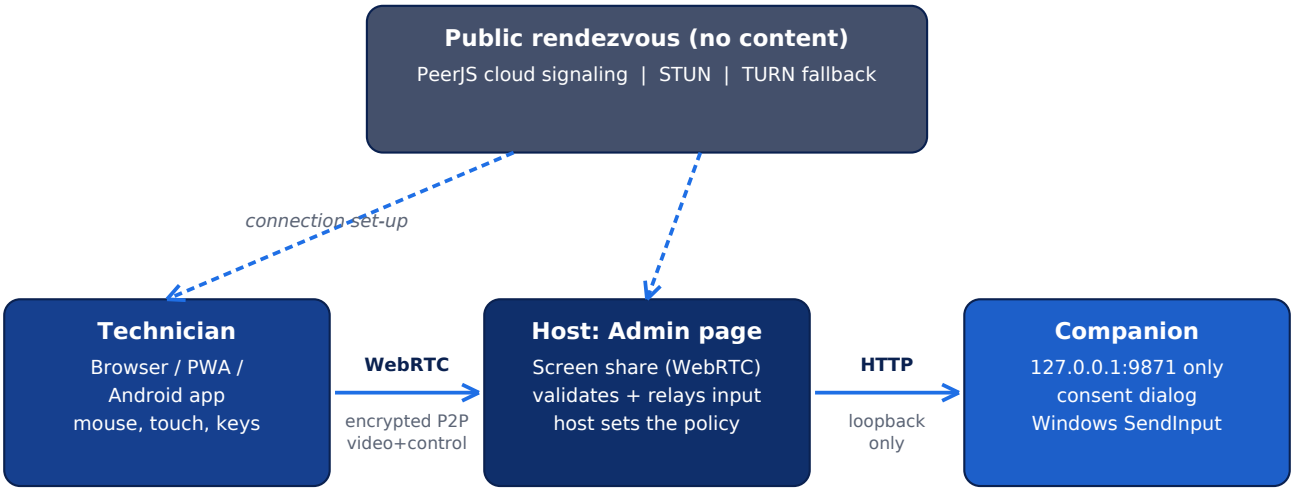


Figure 1 - Components and data paths. Dashed = connection set-up; solid = session traffic.

Connection set-up. The two browsers find each other through the public PeerJS signaling service and traverse NATs with STUN; a public TURN relay is the fallback for restrictive networks. These services see connection metadata only. **Session traffic** - video, control events and tool output - then flows peer to peer, encrypted by WebRTC.

The host page is the policy point. Technician messages arrive at the host's browser, which checks the current access level, validates and sanitises each event, and only then forwards it to the companion on the loopback interface. The companion repeats the validation and refuses to act unless remote control has been explicitly approved by the person at the PC.

4.1 Control protocol

Event	Meaning	Notes
mv / md / mu	Mouse move, button down, button up	Coordinates are fractions (0-1) of the shared screen, so they are independent of resolution and window size
wh	Wheel / scroll	Pixel deltas; clamped; converted to wheel notches
kd / ku	Key down / up	Sent as the physical key code; the companion maps it to a hardware scancode so shortcuts behave as on a real keyboard
tx	Text	Unicode text from on-screen keyboards, injected as Unicode key events (max 200 characters per event)

Events are queued and sent in batches roughly every 8 ms, with consecutive moves coalesced so that slow links never build a backlog of stale mouse positions. The video is letter-boxed inside the viewer, so pointer coordinates are first mapped to the actual picture area before being normalised.

4.2 Touch model

Gesture	Direct touch mode	Touchpad mode
One-finger tap	Left-click where you tap	Left-click at the cursor
Long-press	Right-click where you press	Right-click at the cursor
One-finger drag	Drag (press, move, release)	Moves the cursor (light acceleration)
Two-finger drag	Scroll	Scroll
Two-finger tap	Right-click	Right-click
Double-tap then drag	-	Hold-and-drag (e.g. move a window)

Mouse and pen use the same pointer pipeline. Hardware keyboards are captured while the viewer has focus; a Keys menu sends combinations a browser cannot capture (Win, Alt+Tab, Alt+F4, Task Manager). A hidden text field lets phone keyboards, including non-English input methods, type into the host.

5. Security and privacy

5.1 Controls

Threat	Control
A web page or remote user takes over the PC without consent	Windows shows a native "Allow remote control?" dialog; nothing is injected until the person at the PC clicks Yes. After a refusal the dialog cannot be re-triggered for 8 seconds.
Control continues after the session is forgotten	Control expires after 45 seconds unless the host page sends its 15-second heartbeat; closing the page, ending the share or choosing View Only revokes it immediately and releases all held keys and buttons.
A technician changes the host's access level	The access level is host-authoritative; permission messages from technicians are ignored, and the technician UI cannot change them.
Other websites call the companion	The companion listens on 127.0.0.1 only (LAN exposure is opt-in) and accepts control requests only from localhost and *.github.io origins; the launcher endpoint is gated the same way.
Malformed or hostile input	Events are whitelisted, numerically clamped and size-limited in the host page and again in the companion; request bodies are capped at 64 KB.
Diagnostics abused as a scanning proxy	Only seven allow-listed tools run; each argument is validated against a strict pattern; at most 6 jobs run at once; every request is written to the host's audit log.
Two technicians fight for the mouse	One active controller at a time (hand-off after 3 seconds idle); others are told the PC is busy.
Room overcrowding	A hard cap of 10 technicians; additional peers are rejected, and the host can remove any peer.

5.2 Privacy of the data path

WebRTC media and data channels are encrypted between the two browsers. The public signaling service learns peer identifiers and connection metadata; a TURN relay, if used, forwards already-encrypted packets. No screen content, keystroke or tool result is stored by ShivTrix; the host's audit log lives in the host page.

5.3 Residual risks and recommendations

- **The room link is the key to the room.** Anyone holding it can join as a View-Only technician. Share it privately, use "Change ID" between sessions, and remove unknown peers.
- **No per-technician identity yet.** The host sees an anonymous technician number. Approve control only for people you can identify by another channel (for example a phone call).
- **The approved technician sees everything on screen.** Close sensitive windows first. Consider the screen-capture notice shown by the browser as part of the audit trail.
- **Public infrastructure.** The default PeerJS cloud and OpenRelay TURN are free community services without a service-level guarantee. Organisations should self-host a PeerServer and a TURN server.
- **Unsigned binaries.** The EXE and APK produced by the build kit are unsigned until you add your own certificate/keystore.

6. Deployment models

Model	Use when	Setup
A. Local only	A single PC, no internet hosting	Run companion.bat or ShivTrixAdmin.exe; open http://127.0.0.1:9871
B. Pages + companion	Technicians and hosts use one public address	Publish the web folder to GitHub Pages; hosts also run the companion locally (required for control and tools)
C. Technician on mobile	Support from a phone or tablet	Install the PWA from Chrome (Add to Home screen) or the Android APK
D. Managed fleet	IT department	Push the installer with your deployment tool; self-host signaling/TURN; sign the binaries

Why a host companion is unavoidable. Browsers deliberately cannot inject input into the operating system or run native network tools. The companion is the minimal, inspectable bridge: a single Python file (also packaged as an EXE) bound to the loopback interface.

7. Quality assurance

The release was verified with an automated end-to-end suite that drives two real browser pages against the real companion, using a simulated peer network. All 24 checks pass:

Area	Checks
Access control	Technician cannot raise or forge the access level; input is rejected until control is enabled; control is withdrawn on revoke; companion returns to idle
Mouse and keyboard	Move, click, right-click, wheel, Ctrl+C and typed text reach the host bridge; clicks map to the correct point on the host screen; Win+D key combination
Touch	Tap, long-press, drag and two-finger scroll in Direct mode; relative cursor movement in Touchpad mode; on-screen-keyboard Unicode text and Backspace
Tools on host	Port check and adapter inventory executed on the host through the peer link and logged in the host audit log; target switch disabled when control is revoked
Scale and stability	10-peer cap enforced and the eleventh peer told the room is full; no JavaScript errors on either page
PWA	Manifest and icons load, service worker active, offline reload works, shortcuts open the correct view

Not yet verified in this release

Real Windows input injection (the SendInput structures were checked for the correct 40-byte layout but not executed on Windows), behaviour over the public PeerJS/TURN services and real-world latency, and the packaged EXE and APK (their build scripts and cloud workflows are provided but were not executed). Run the pilot checklist in the SOP before relying on the product in production.

8. Limitations and roadmap

Today	Planned
Windows hosts only for control (Linux/macOS hosts can share and run tools)	macOS and Linux input engines
Phones cannot host a screen share (mobile browsers do not allow it)	Native Android host using the MediaProjection API
Controls the primary display by default; choice of display is manual	Automatic display detection
Attended sessions only	Unattended access with a Windows service and device pairing
No clipboard or file transfer	Clipboard sync and drag-and-drop file transfer
Windows will not accept input for administrator-level windows unless the Elevated build is used	Optional service mode for the secure desktop
Quick-trigger buttons "Flush DNS" and "Defender Scan" only log a request	Signed, allow-listed maintenance actions
Anonymous technician numbers	Technician accounts, session recording and exportable audit trail

9. Conclusion

ShivTrix Admin shows that a remote-assistance experience close to a dedicated desktop tool can be delivered through the browser, provided the host runs a small, transparent companion and the design keeps the human at the PC in charge. Its distinguishing features are touch-first control, diagnostics that run where the problem is, and a consent model that is enforced by the operating system rather than by a web page. The next steps for production use are signed binaries, self-hosted rendezvous services, technician identity, and a pilot on real networks.

Appendix A - Message reference

Message	Direction	Purpose
permission_change	Host to technicians	Announces the access level (ignored if sent by a technician)
ctl_state	Host to technicians	Control approved / withdrawn by the person at the PC
ctl	Technician to host	Batch of up to 100 input events
ctl_denied	Host to technician	Control not enabled, or another technician is active
tool_req / tool_cancel	Technician to host	Start / stop an allow-listed diagnostic
tool_start / tool_chunk / tool_end / tool_error	Host to technician	Streams the diagnostic output back
rtt_ping / rtt_pong	Both	Latency indicator shown in the control toolbar
room_full / kicked	Host to technician	Peer limit reached / removed by the host

Appendix B - Companion endpoints

Endpoint	Purpose
GET /api/control/status	Control state (idle, pending, armed, denied), displays
POST /api/control/arm | disarm | monitor	Request approval (shows the dialog) / revoke / choose display
POST /api/control/input	Inject a validated batch of events (only while approved)
GET /api/powertoys/*	health, portcheck, ping, ping-stream, traceroute-stream, nslookup, adapters
GET /metrics	CPU / memory telemetry

Appendix C - Glossary

Term	Meaning
PWA	Progressive Web App: a website that installs like an app and works offline
WebRTC	Browser technology for encrypted peer-to-peer video, audio and data
STUN / TURN	Services that help two peers connect through routers and firewalls (TURN relays traffic as a last resort)
SendInput	The Windows API for injecting mouse and keyboard input
Host / Technician	The computer being helped / the person helping